Saksham Sharma

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Education

Thapar Institute of Engineering and Technology — Patiala, Punjab
Bachelor of Technology in Electronics and Computer Engineering — *Aug 2023 – May 2027*

Projects

Agentic Coding Agent — 2026
Python, LangGraph, Docker, FastAPI, PostgreSQL, GCP

- Raised the pass rate on problems built to defeat memorized answers from 67% to 100% (12/12 runs) by feeding failing test results back into a self-correcting retry loop capped at 3 retries.
- Achieved a 100% pass rate across 18 algorithm problems (54/54 runs) at ∼1K tokens and 3.4 s per run, graded against tests 88% hidden from the model, by building an agent that plans, writes and verifies Python in a Docker sandbox.
- Ran untrusted AI-written code publicly on a 1 GB free-tier VM with 0 out-of-memory crashes (peak 670 MB) by isolating each run in a network-less, resource-capped container and limiting each IP to 5 runs/day before any model call.

CI/CD Pipeline with Automated Rollback — 2026
GitHub Actions, Docker, Kubernetes

- Built a CI/CD pipeline (GitHub Actions, Docker, Kubernetes) with automated rollback to ship containerized services safely.
- Verified the rollback path in a live failure drill with zero dropped requests.

URL Shortener API with Click Analytics — July 2026
Python, FastAPI, PostgreSQL, SQLAlchemy, pytest

- Engineered a production-grade REST API with collision-free short-code generation, real-time click analytics, and async SQLAlchemy; deployed on Railway with PostgreSQL.
- Built a 13-test suite with pytest + pytest-asyncio covering core functionality, connection pooling, and edge cases.
- Designed a normalized PostgreSQL schema with indexed columns, foreign-key constraints, and Alembic migrations; prevented SQL injection via parameterized queries.

Technical Skills

Languages: Python, SQL, C/C++
Backend: FastAPI, REST APIs, async/await
Databases: PostgreSQL, SQLAlchemy
Infra/Tools: Docker, Kubernetes, GitHub Actions, GCP, Git, pytest
AI: LangGraph, Gemini API
Core CS: Data Structures, Algorithms, OOP, System Design

Leadership and Extracurricular

Vice President, TED-Ed Club – Spring Dale Senior School (1-year term)
Participant, Smart India Hackathon (SIH)